

Matrix Arithmetic

Let S be one of the sets $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ or $\mathbb{C}$.

- **Definition 1**

Let $m, n \in \mathbb{N}$. A **matrix** of size $m \times n$ with entries in S is a rectangular array of the form

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m \times n}, \quad a_{ij} \in S \text{ for } 1 \leq i \leq m, 1 \leq j \leq n.$$

where m is the number of rows and n is the number of columns.

The set of all such matrices is denoted by $M_{m \times n}(S)$.

The i -th row of A is

$$R_i = [a_{i1} \quad a_{i2} \quad \cdots \quad a_{in}].$$

The j -th column of A is

$$C_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

The element of A that is in the i -th row and the j -th column is denoted a_{ij} and we can write

$$A = [a_{ij}]_{m \times n}.$$

- **Definition 2**

Two matrices $A = [a_{ij}]_{m \times n}, B = [b_{ij}]_{m \times n} \in M_{m \times n}(S)$ are **equal** if $a_{ij} = b_{ij}$ for all $1 \leq i \leq m, 1 \leq j \leq n$.

- **Definition 3**

A matrix A is said to be **square** if the number of rows is equal to the number of columns, that is,

$$A = [a_{ij}]_{n \times n}.$$

- **Notation 3.1**

$$[a_{ij}]_{n \times n} = [a_{ij}]_n, \quad M_{n \times n}(S) = M_n(S).$$

- **Definition 4**

The **zero matrix** is the matrix whose entries are all 0 and is denoted $\mathbb{0}$.

- **Example 4.1**

$$\mathbb{0}_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbb{0}_{4 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbb{0}_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

- **Definition 5**

Given two matrices $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n} \in M_{m \times n}(S)$, we define the sum as the matrix $A + B = [a_{ij} + b_{ij}]_{m \times n}$.

◦ **Example 5.1**

Let

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 7 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -2 & 3 & 1 \\ 4 & -2 & 1 \end{bmatrix}.$$

Then we have the sum

$$A + B = \begin{bmatrix} 1 + (-2) & -2 + 3 & 0 + 1 \\ 3 + 4 & 7 + (-2) & 1 + 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 7 & 5 & 2 \end{bmatrix}.$$

• **Definition 6**

Given a matrix $A = [a_{ij}]_{m \times n} \in M_{m \times n}(S)$ and $\alpha \in S$, we define the **scalar product** as the matrix

$$\alpha A = [\alpha a_{ij}]_{m \times n}.$$

◦ **Example 6.1**

If

$$A = \begin{bmatrix} 1 & 7 \\ 2 & 6 \\ 3 & 5 \end{bmatrix},$$

then we have

$$2A = \begin{bmatrix} 2 & 14 \\ 4 & 12 \\ 6 & 10 \end{bmatrix}.$$

If

$$A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix},$$

then we have

$$iA = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

◦ **Theorem 6.2**

Let $A, B, C \in M_{m \times n}(S)$ and $\alpha, \beta \in S$, then we have the following properties

1. $(A + B) + C = A + (B + C)$.
2. $A + B = B + A$.
3. $A + \mathbb{0} = A$.
4. $A + (-A) = \mathbb{0}$.
5. $(\alpha + \beta)A = \alpha A + \beta A$.
6. $\alpha(A + B) = \alpha A + \alpha B$.
7. $(\alpha\beta)A = \alpha(\beta A)$.

■ **Proof 6.2.1**

Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$, $C = [c_{ij}]_{m \times n}$.

$$\begin{aligned}
1. \quad (A + B) + C &= [a_{ij} + b_{ij}]_{m \times n} + [c_{ij}]_{m \times n} \\
&= [(a_{ij} + b_{ij}) + c_{ij}]_{m \times n} \\
&= [a_{ij} + (b_{ij} + c_{ij})]_{m \times n} \\
&= [a_{ij}]_{m \times n} + [b_{ij} + c_{ij}]_{m \times n} \\
&= A + (B + C).
\end{aligned}$$

$$\begin{aligned}
2. \quad A + B &= [a_{ij} + b_{ij}]_{m \times n} \\
&= [b_{ij} + a_{ij}]_{m \times n} \\
&= B + A.
\end{aligned}$$

$$\begin{aligned}
3. \quad A + \mathbf{0} &= [a_{ij} + 0]_{m \times n} \\
&= [a_{ij}]_{m \times n} \\
&= A.
\end{aligned}$$

$$\begin{aligned}
4. \quad A + (-A) &= [a_{ij}]_{m \times n} + [-a_{ij}]_{m \times n} \\
&= [a_{ij} + (-a_{ij})]_{m \times n} \\
&= [0]_{m \times n} \\
&= \mathbf{0}.
\end{aligned}$$

$$\begin{aligned}
5. \quad (\alpha + \beta)A &= [(\alpha + \beta)a_{ij}]_{m \times n} \\
&= [\alpha a_{ij} + \beta a_{ij}]_{m \times n} \\
&= [\alpha a_{ij}]_{m \times n} + [\beta a_{ij}]_{m \times n} \\
&= \alpha A + \beta A.
\end{aligned}$$

$$\begin{aligned}
6. \quad \alpha(A + B) &= \alpha[a_{ij} + b_{ij}]_{m \times n} \\
&= [\alpha(a_{ij} + b_{ij})]_{m \times n} \\
&= [\alpha a_{ij} + \alpha b_{ij}]_{m \times n} \\
&= [\alpha a_{ij}]_{m \times n} + [\alpha b_{ij}]_{m \times n} \\
&= \alpha A + \alpha B.
\end{aligned}$$

$$\begin{aligned}
7. \quad (\alpha\beta)A &= [(\alpha\beta)a_{ij}]_{m \times n} \\
&= [\alpha(\beta a_{ij})]_{m \times n} \\
&= \alpha[\beta a_{ij}]_{m \times n} \\
&= \alpha(\beta A).
\end{aligned}$$

• **Definition 7**

Consider two matrices $A = [a_{ij}]_{m \times n} \in M_{m \times n}(S)$ and $B = [b_{jk}]_{n \times p} \in M_{n \times p}(S)$. We define the **product** as the matrix

$$C = [c_{ik}]_{m \times p}$$

where

$$c_{ik} = \sum_{j=1}^n a_{ij}b_{jk}.$$

◦ **Remark 7.1**

For the product between A and B to make sense, the number of columns of A must be equal to the number of rows of B .

◦ **Example 7.2**

Let

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} -2 & 5 \\ 4 & -3 \\ 2 & 1 \end{bmatrix}.$$

Then by the definition of the matrix product, the resulting matrix AB has size 2×2 and can be written as

$$AB = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

where each entry c_{ik} is computed explicitly by taking the dot product of the i -th row of A and the k -th column of B as follows

$$c_{11} = \sum_{j=1}^3 a_{1j}b_{j1} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} = 1 \cdot (-2) + 2 \cdot 4 + (-1) \cdot 2 = -2 + 8 - 2 = 4.$$

$$c_{12} = \sum_{j=1}^3 a_{1j}b_{j2} = a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} = 1 \cdot 5 + 2 \cdot (-3) + (-1) \cdot 1 = 5 - 6 - 1 = -2.$$

$$c_{21} = \sum_{j=1}^3 a_{2j}b_{j1} = a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} = 3 \cdot (-2) + 1 \cdot 4 + 4 \cdot 2 = -6 + 4 + 8 = 6.$$

$$c_{22} = \sum_{j=1}^3 a_{2j}b_{j2} = a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} = 3 \cdot 5 + 1 \cdot (-3) + 4 \cdot 1 = 15 - 3 + 4 = 16.$$

Therefore, substituting these calculated entries back into the product matrix yields the final result

$$AB = \begin{bmatrix} 4 & -2 \\ 6 & 16 \end{bmatrix}.$$

◦ **Theorem 7.3**

Let A, B, C be matrices with sizes such that the following operations make sense and let $\alpha \in S$. Then

1. $(AB)C = A(BC)$.
2. $(A + B)C = AC + BC$.
3. $A(B + C) = AB + AC$.
4. $\alpha AB = (\alpha A)B = A(\alpha B)$.

■ **Proof 7.3.1**

1. Let $A = [a_{ij}]_{m \times n}$, $B = [b_{jk}]_{n \times p}$, $C = [c_{kl}]_{p \times q}$. Then

$$\begin{aligned} (AB)C &= [d_{ik}]_{m \times p} [c_{kl}]_{p \times q} \\ &= [e_{il}]_{m \times q} \end{aligned}$$

◦ **Theorem 7.3**

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■ **Proof 7.3.1**

1. Let $A = [a_{ij}]_{m \times n}$, $B = [b_{jk}]_{n \times p}$ and $C = [c_{kl}]_{p \times q}$. Then

$$\begin{aligned}
(AB)C &= ([a_{ij}]_{m \times n} [b_{jk}]_{n \times p}) [c_{kl}]_{p \times q} \\
&= [\sum_{j=1}^n a_{ij} b_{jk}]_{m \times p} [c_{kl}]_{p \times q} \\
&= [\sum_{k=1}^p (\sum_{j=1}^n a_{ij} b_{jk}) c_{kl}]_{m \times q} \\
&= [\sum_{k=1}^p \sum_{j=1}^n a_{ij} b_{jk} c_{kl}]_{m \times q} \\
&= [\sum_{j=1}^n \sum_{k=1}^p a_{ij} b_{jk} c_{kl}]_{m \times q} \\
&= [\sum_{j=1}^n a_{ij} (\sum_{k=1}^p b_{jk} c_{kl})]_{m \times q} \\
&= [a_{ij}]_{m \times n} [\sum_{k=1}^p b_{jk} c_{kl}]_{n \times q} \\
&= [a_{ij}]_{m \times n} ([b_{jk}]_{n \times p} [c_{kl}]_{p \times q}) \\
&= A(BC).
\end{aligned}$$

2. Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$, and $C = [c_{jk}]_{n \times p}$. Then

$$\begin{aligned}
(A+B)C &= ([a_{ij}]_{m \times n} + [b_{ij}]_{m \times n}) [c_{jk}]_{n \times p} \\
&= [a_{ij} + b_{ij}]_{m \times n} [c_{jk}]_{n \times p} \\
&= [\sum_{j=1}^n (a_{ij} + b_{ij}) c_{jk}]_{m \times p} \\
&= [\sum_{j=1}^n (a_{ij} c_{jk} + b_{ij} c_{jk})]_{m \times p} \\
&= [\sum_{j=1}^n a_{ij} c_{jk} + \sum_{j=1}^n b_{ij} c_{jk}]_{m \times p} \\
&= [\sum_{j=1}^n a_{ij} c_{jk}]_{m \times p} + [\sum_{j=1}^n b_{ij} c_{jk}]_{m \times p} \\
&= [a_{ij}]_{m \times n} [c_{jk}]_{n \times p} + [b_{ij}]_{m \times n} [c_{jk}]_{n \times p} \\
&= AC + BC.
\end{aligned}$$

3. Let $A = [a_{ij}]_{m \times n}$, $B = [b_{jk}]_{n \times p}$, and $C = [c_{jk}]_{n \times p}$. Then

$$\begin{aligned}
A(B+C) &= [a_{ij}]_{m \times n} ([b_{jk}]_{n \times p} + [c_{jk}]_{n \times p}) \\
&= [a_{ij}]_{m \times n} [b_{jk} + c_{jk}]_{n \times p} \\
&= [\sum_{j=1}^n a_{ij} (b_{jk} + c_{jk})]_{m \times p} \\
&= [\sum_{j=1}^n (a_{ij} b_{jk} + a_{ij} c_{jk})]_{m \times p} \\
&= [\sum_{j=1}^n a_{ij} b_{jk} + \sum_{j=1}^n a_{ij} c_{jk}]_{m \times p} \\
&= [\sum_{j=1}^n a_{ij} b_{jk}]_{m \times p} + [\sum_{j=1}^n a_{ij} c_{jk}]_{m \times p} \\
&= [a_{ij}]_{m \times n} [b_{jk}]_{n \times p} + [a_{ij}]_{m \times n} [c_{jk}]_{n \times p} \\
&= AB + AC.
\end{aligned}$$

4. Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{jk}]_{n \times p}$. Then

$$\begin{aligned}
\alpha(AB) &= \alpha([a_{ij}]_{m \times n} [b_{jk}]_{n \times p}) \\
&= \alpha[\sum_{j=1}^n a_{ij} b_{jk}]_{m \times p} \\
&= [\alpha \sum_{j=1}^n a_{ij} b_{jk}]_{m \times p} \\
&= [\sum_{j=1}^n \alpha(a_{ij} b_{jk})]_{m \times p} \\
&= [\sum_{j=1}^n (\alpha a_{ij}) b_{jk}]_{m \times p} = [\sum_{j=1}^n a_{ij} (\alpha b_{jk})]_{m \times p} \\
&= [\alpha a_{ij}]_{m \times n} [b_{jk}]_{n \times p} = [a_{ij}]_{m \times n} [\alpha b_{jk}]_{n \times p} \\
&= (\alpha A)B = A(\alpha B).
\end{aligned}$$

◦ **Remark 7.4**

The product of matrices is not commutative. For example, consider

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 4 & -4 \end{bmatrix}$$
$$BA = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 7 \\ 4 & -2 \end{bmatrix}$$

That is, $AB \neq BA$.

◦ **Remark 7.5**

The product between two nonzero square matrices can be zero matrix. For example

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$